KNOWLEDGE ENGINEERING RESEARCH (GROUP OUTPUT)

Project: Diet & Nutrition Expert System
Role: Knowledge Engineering Team
Course: DCIT 313 – Artificial Intelligence

1. SOURCES USED

Our knowledge was obtained from:

- World Health Organization (WHO)
- NIH – Office of Dietary Supplements
- CDC
- USDA FoodData Central
- Peer-reviewed nutrition literature

All information is evidence-based and medically verified.

2. SYSTEM INPUTS (WHAT THE MODEL RECEIVES)

The expert system receives the following inputs from the user:

- **Age** (years)
- **Height** (meters or cm)
- **Weight** (kg)
- **Health Goal** (one of):
  - weight_loss
  - weight_gain
  - muscle_gain
  - maintain
- **Symptoms** (multiple selection):
  - fatigue
  - constipation

- headache
- muscle_cramps
- poor_concentration

- **Allergies / Food Restrictions** (optional):
  - dairy
  - nuts
  - eggs
  - gluten
  - shellfish

2. BALANCED DIET KNOWLEDGE (Preventive Health Mode)

Age-Based Nutritional Requirements

Children (2–12 years)

Component	Requirement
Energy	1400–2200 kcal/day (depends on age and activity level)
Protein	0.95 g per kg body weight/day
Carbohydrates	45–65% of total calories
Fat	25–35% of total calories
Fibre	19–25 g/day
Water	1.5–2 litres/day

Special considerations:

• Calcium and vitamin D are essential for bone growth.
• Iron supports brain development and oxygen transport.
• Balanced meals support growth and immune system development.

Adolescents (13–18 years)

Component	Requirement
Energy	2000–3200 kcal/day depending on sex and activity
Protein	0.85 g per kg body weight/day
Carbohydrates	45–65% of total calories
Fat	25–35% of total calories
Fibre	22–31 g/day
Water	2–3 litres/day

Special considerations:

• Higher iron requirements especially for teenage girls.
• Calcium intake is critical for peak bone development.
• Adequate calories are needed to support rapid growth and hormonal changes.

Adults (19–50 years)

Component	Requirement
Energy	2000–2500 kcal/day (depends on activity)
Protein	0.8 g per kg body weight/day
Carbohydrates	45–65% of total calories
Fat	20–35% of total calories
Fibre	25–38 g/day
Water	2–3 litres/day

Special considerations:

• Increased calcium and vitamin D are needed for bone health.
• Iron needs are higher in women due to menstruation.

Older Adults (Above 50 years)

Component	Requirement
Energy	1800–2200 kcal/day depending on activity
Protein	1.0–1.2 g per kg body weight/day
Carbohydrates	45–65% of total calories
Fat	20–35% of total calories
Fibre	21–30 g/day
Water	2–2.5 litres/day

Special considerations:

• Higher vitamin D and calcium intake is needed to prevent bone loss.
• Vitamin B12 absorption may decrease with age.
• Protein intake is important to prevent muscle loss.

Macronutrient Distribution

A balanced diet should contain:

• Carbohydrates: 45–65%
• Protein: 10–35%
• Fat: 20–35%
• Fibre: minimum 25 g/day

These values are used by the system to guide diet plans for:

• Weight loss
• Weight gain
• Muscle gain
• Maintenance

Micronutrient Requirements

Nutrient	Function	Deficiency Effects	Daily Need
Iron	Oxygen transport	Fatigue, anemia	8–18 mg
Calcium	Bones & teeth	Weak bones	1000 mg
Vitamin D	Calcium absorption	Bone pain	600–800 IU
Vitamin B12	Nerve & blood cells	Anemia, memory loss	2.4 mcg
Zinc	Immune system	Frequent infections	8–11 mg
Magnesium	Muscle & nerve function	Cramps	310–420 mg

Allergy & Restriction Handling

Common allergens:

• Dairy
• Nuts
• Eggs
• Gluten
• Shellfish

Substitution rules:

• Dairy → soy milk, almond milk
• Wheat → rice, maize
• Nuts → seeds or legumes

Cross-contamination risk is noted for severe allergies.

Food Composition Knowledge

Whole foods:

• Fruits
• Vegetables
• Whole grains
• Legumes

Processed foods:

• Sugary drinks
• Refined flour
• Packaged snacks

Example:

Spinach is a whole food rich in iron and vitamin C.
Oats are high in fibre and complex carbohydrates.

3. DERIVED VALUE: BODY MASS INDEX (BMI)

Meaning of BMI

Body Mass Index (BMI) is a numerical value calculated from a person's weight and height and is used to classify their weight status.

BMI Formula

$$BMI = \frac{\text{Weight (kg)}}{\text{Height (m)}^2}$$

BMI Categories Used by the System

BMI Value	Category
< 18.5	Underweight
18.5 – 24.9	Normal weight
≥ 25	Overweight

BMI is used only to **guide nutrition advice** and not to diagnose disease.

The BMI categories used in this system are based on World Health Organization (WHO) standards for adult body mass index classification.

System Age Policy (For Your Knowledge Base)

Case 1: Age < 19

System rule:

IF age < 19

THEN

- Do NOT classify using adult BMI categories
- Display message:

  "BMI interpretation for children and adolescents requires growth charts.

  Please consult a healthcare professional."
- Provide only general healthy eating advice

Case 2: Age 19 – 50

Use standard WHO BMI categories:

<18.5 → Underweight

18.5–24.9 → Normal

≥25 → Overweight

Case 3: Age > 50

System rule:

IF age > 50

THEN

- Use BMI cautiously
- Provide nutrition advice
- Include note:

  "BMI interpretation in older adults may vary. Consult a healthcare professional for personalized assessment."

BMI Interpretation Rules (Final)

IF age < 19

THEN

no BMI category

show warning message

IF 19 ≤ age ≤ 50:

IF BMI < 18.5 → Underweight

IF 18.5 ≤ BMI < 25 → Normal

IF BMI ≥ 25 → Overweight

IF age > 50:

use adult BMI

but show caution message

4. DECISION LOGIC (HOW THE SYSTEM THINKS)

The expert system follows this order:

1. Accept user inputs
2. Calculate BMI
3. Determine BMI category
4. Apply goal-based diet rules
5. Apply symptom-based rules
6. Filter foods based on allergies
7. Generate structured output
8. Display disclaimer

5. SYMPTOM-BASED NUTRITION KNOWLEDGE (Support Mode)

For each symptom, we define:

1. Possible nutritional cause
2. Scientific explanation
3. Recommended foods
4. Safety rule

Symptom: Fatigue

Possible Causes:

- Iron deficiency
- Low blood sugar
- Dehydration

Scientific Explanation:

Iron is needed to form hemoglobin, which carries oxygen in the blood. Low iron reduces oxygen supply to tissues, causing tiredness.

Recommended Foods:

- Spinach
- Beans
- Red meat
- Fortified cereals
- Oranges (improves iron absorption)

Safety Rule:

If fatigue lasts more than 2 weeks or includes dizziness, the system advises medical consultation.

Symptom: Constipation

Possible Causes:

- Low fibre intake

- Low water intake

Scientific Explanation:

Fibre adds bulk to stool and stimulates bowel movement. Without enough fibre, stool becomes hard and difficult to pass.

Recommended Foods:

- Oats
- Fruits (papaya, apples)
- Vegetables
- Whole grains

Safety Rule:

If constipation lasts more than 7 days or includes blood in stool, the system advises medical attention.

Symptom: Headache

Possible Causes:

- Dehydration
- Low magnesium
- Low blood sugar

Scientific Explanation:

Dehydration reduces blood volume and oxygen delivery to the brain, causing headaches.

Recommended Foods:

- Water
- Bananas
- Nuts
- Whole grains

Safety Rule:

Severe or frequent headaches trigger medical referral advice.

Symptom: Muscle Cramps

Possible Causes:

- Low magnesium
- Low potassium
- Dehydration

Scientific Explanation:

Electrolytes regulate muscle contraction. Deficiency leads to abnormal muscle contraction.

Recommended Foods:

- Bananas
- Leafy greens
- Beans
- Milk

Safety Rule:

Persistent cramps → seek professional care.

Symptom: Poor Concentration

Possible Causes:

- Vitamin B12 deficiency
- Low glucose
- Iron deficiency

Scientific Explanation:

B12 is required for nerve function and red blood cell production. Low levels affect brain oxygen supply and focus.

Recommended Foods:

- Eggs
- Fish
- Meat
- Fortified cereals

Safety Rule:

Long-term cognitive problems → medical advice.

6. KNOWLEDGE REPRESENTATION (RULE FORM)

Examples:

IF BMI < 18.5
THEN classify = underweight

IF symptom = fatigue
AND cause = iron deficiency
THEN recommend iron-rich foods

IF goal = weight loss
THEN recommend high fibre + lean protein

IF allergy = dairy
THEN substitute with plant-based milk

7. OUTPUT STRUCTURE

System outputs:

- BMI value
- BMI category
- Goal-based advice
- Symptom-based advice
- Food recommendations

- Disclaimer

8. ETHICAL RULES

Our group ensured that:

- The system does NOT diagnose disease
- The system does NOT prescribe medication
- All advice is general nutrition guidance
- Severe symptoms lead to medical referral
- A disclaimer is included

9. SUMMARY (GROUP CONTRIBUTION)

As the Knowledge Engineering team, we:

- Collected scientific nutrition knowledge
- Defined nutrient requirements
- Mapped symptoms to causes and foods
- Created structured IF–THEN rules
- Ensured safety and ethics
- Defined inputs
- Defined BMI and its use
- Built safety logic

Our work provides the scientific foundation for the expert system.